Absence of pomegranate ellagitannins in the majority of commercial Pomegranate extracts: implications for standardization and quality control.

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The health benefits associated with pomegranate juice have led to the development of pomegranate extracts as botanical dietary supplements. Pomegranates contain hydrolyzable tannins in the form of punicalagins and punicalin as well as tannin-based complex oligomers that account for much of the antioxidant activity in juice. The content of ellagic acid has been used to standardize most pomegranate extract dietary supplements marketed. However, supplements can be adulterated with ellagic acid from less expensive plant sources and undercut this method of standardization. To compare the phytochemical contents and antioxidant activities of commercially available pomegranate extract dietary supplements beyond their content of ellagic acid, a total of 27 different supplements in the form of capsules, tablets, and soft gels were studied. Total phenolics were measured using both gallic acid equivalent (GAE) and ellagic acid equivalent (EAE) assays. Punicalagins, punicalin, and ellagic acid contents were determined by HPLC, whereas antioxidant capacity was measured using the Trolox equivalent antioxidant capacity (TEAC) assay. Of the 27 supplements tested, only 5 had the typical pomegranate tannin profile by HPLC, 17 had ellagic acid as the predominant chemical with minor or no detectable pomegranate tannins, and 5 had no detectable tannins or ellagic acid. Therefore, standardization of pomegranate extract supplements based on their ellagic acid content does not guarantee pomegranate supplement authenticity. Future research is needed to assess the health impact of substituting ellagic acid for the complex mix of phytochemicals in a pomegranate extract dietary supplement.